

VENANGO, PA, USA

WMO: 725267

Lat: 41.383N

Lon: 79.867W

Elev: 469

StdP: 95.81

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94868

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-14.1	-22.0	0.6	-16.0	-18.8	0.7	-13.1	10.4	1.8	9.2	0.2	3.3	240	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	29.8	22.1	28.1	21.0	27.4	20.4	23.4	27.8	22.5	26.7	21.6	25.6	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.9	26.2	21.2	16.8	25.0	20.2	15.8	23.9	71.9	27.6	68.5	26.3	65.3	25.3	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.7	7.7	6.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-19.8	31.7	3.5	1.6	-22.4	32.9	-24.4	33.8	-26.4	34.7	-29.0	35.9		
			-20.2	25.0	3.4	0.9	-22.7	25.6	-24.7	26.2	-26.7	26.7	-29.2	27.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-3.8	-2.7	1.9	8.6	14.3	18.6	20.8	20.0	16.5	10.3	4.3	-0.8
	DBStd	10.02	6.72	6.41	6.34	5.42	4.70	3.54	2.91	2.80	3.86	4.70	5.10	5.36
	HDD10.0	1728	430	356	264	89	14	0	0	0	2	54	183	337
	HDD18.3	3633	687	588	510	295	141	39	10	16	81	252	422	593
	CDD10.0	1388	2	2	13	47	147	258	335	311	197	64	11	2
	CDD18.3	251	0	0	0	3	17	48	87	67	26	3	0	0
	CDH23.3	1810	0	0	2	31	154	345	632	448	180	18	0	0
CDH26.7	387	0	0	0	6	24	74	152	94	34	2	0	0	
Wind	WSAvg	3.1	3.8	3.6	3.5	3.6	3.1	2.7	2.4	2.2	2.5	3.1	3.5	3.6
Precipitation	PrecAvg	1097	83	66	85	93	99	101	121	104	103	87	85	86
	PrecMax	1401	169	149	161	151	171	175	257	195	207	192	143	171
	PrecMin	815	38	35	35	44	35	24	32	41	29	36	36	30
	PrecStd	159	32	30	35	25	36	42	64	36	51	34	30	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	17.1	22.4	27.1	28.9	30.9	32.2	31.2	29.7	25.7	19.1	16.0
		MCWB	12.1	11.4	14.3	16.5	20.3	22.6	23.6	22.1	21.6	19.5	13.5	12.1
	2%	DB	12.4	12.5	17.9	23.0	27.3	28.8	29.8	28.9	27.5	22.6	17.2	12.7
		MCWB	10.8	9.3	11.4	14.3	18.8	21.4	22.7	21.6	20.5	16.7	12.2	10.5
	5%	DB	9.0	9.1	15.2	21.0	25.1	27.3	28.0	27.5	25.8	20.9	14.9	10.0
		MCWB	7.2	6.3	10.1	13.4	17.5	20.5	21.3	20.7	19.5	16.1	11.0	7.7
	10%	DB	6.0	6.4	12.3	17.7	22.7	25.4	27.2	26.2	23.0	17.9	12.6	7.2
		MCWB	3.7	3.3	8.2	11.2	16.3	19.1	20.8	19.8	18.1	14.1	9.4	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	12.8	15.0	18.0	21.8	23.5	25.1	24.1	23.1	20.8	15.2	13.6
		MCDB	15.2	15.5	20.6	24.2	26.7	28.2	29.8	28.7	27.5	24.4	18.0	15.2
	2%	WB	10.9	10.0	13.0	15.8	20.2	22.4	23.7	22.9	21.6	18.7	13.3	10.8
		MCDB	12.4	12.2	16.8	20.9	25.4	27.4	28.2	27.0	25.8	21.5	15.7	12.3
	5%	WB	7.5	6.8	10.9	14.1	18.9	21.4	22.6	21.9	20.4	16.6	11.7	7.9
		MCDB	8.7	9.0	14.6	18.9	23.5	26.0	26.8	25.5	23.9	19.3	14.2	9.1
	10%	WB	3.6	3.8	8.0	12.5	17.5	20.2	21.6	21.0	19.3	14.8	9.8	5.2
		MCDB	5.4	6.2	11.4	16.8	21.5	24.0	25.4	24.4	22.3	17.8	13.0	7.4
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.5	9.5	11.4	11.3	10.6	10.7	10.5	10.5	9.6	7.9	6.5
		MCDBR	9.9	11.9	13.7	15.0	13.6	12.2	11.7	12.0	12.7	12.8	11.5	9.8
		MCWBR	8.7	9.4	8.7	8.8	7.1	6.5	5.8	6.0	6.4	7.6	8.2	8.5
	5% WB	MCDBR	9.7	11.3	12.2	12.4	11.5	10.5	10.0	9.8	10.1	10.6	9.9	9.2
		MCWBR	9.2	9.8	8.6	8.4	6.9	6.2	5.5	5.7	5.6	7.3	8.0	8.8
Clear-Sky Solar Irradiance	taub	0.293	0.323	0.340	0.368	0.404	0.436	0.434	0.425	0.374	0.350	0.310	0.293	
	taud	2.438	2.334	2.380	2.349	2.257	2.198	2.221	2.237	2.361	2.433	2.523	2.495	
	Ebn at Noon	864	882	908	903	873	842	840	836	860	842	840	837	
	Edh at Noon	77	100	108	120	135	143	139	132	109	90	72	68	
All-Sky Solar Radiation	RadAvg	1.40	2.23	3.21	4.40	5.17	5.65	5.74	5.13	4.13	2.66	1.74	1.10	
	RadStd	0.12	0.29	0.35	0.47	0.49	0.49	0.44	0.29	0.43	0.27	0.24	0.11	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	N/A

Nomenclature: See separate page